
# Bangkok Airbnb Market Intelligence Report

## Expernetic Data Engineer Intern Assessment

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# 1. Executive Summary

This report presents a comprehensive market intelligence analysis of the Bangkok short-term rental market using publicly available Inside Airbnb data. The analysis was conducted as part of the Expernetic Data Engineer Intern Technical Assessment, with the objective of transforming raw Airbnb listing data into actionable business insights through end-to-end data engineering, statistical analysis, machine learning, and AI experimentation.

**Dataset:** Bangkok, Thailand - 28,806 listings, 10,514,202 calendar records, and 583,333 guest reviews scraped in September 2025.

**Key Findings:**

- The Bangkok Airbnb market is dominated by entire home/apartment listings (71%), with an average nightly price of 2,141 THB and average occupancy rate of 23.8%.
- Vadhana is the highest-priced neighbourhood at 3,235 THB average nightly rate, commanding an 87% price premium over budget neighbourhoods like Huai Khwang.
- Superhosts achieve significantly higher occupancy rates (30.43%) compared to regular hosts (20.21%), though price differences are negligible - suggesting superhost status drives demand rather than pricing power.
- Sentiment analysis of 10,000 guest reviews revealed 81.96% positive sentiment, though a declining trend from 0.85 (2011) to 0.52 (2024) indicates guests are becoming increasingly critical as the platform matures.
- A Random Forest price prediction model achieved MAE of 866 THB with $R^2$ of 0.49, identifying bedrooms (864 THB SHAP impact) and host tenure as the strongest price drivers.

**Recommendations:**

Hosts seeking maximum revenue should target Vadhana or Parthum Wan neighbourhoods, list entire home properties, and invest in building platform tenure. The data strongly suggests that guest experience quality - not pricing strategy, is the primary driver of long-term booking success in Bangkok.

# 2. Objectives & Scope

## 2.1 Project Objectives

This assessment was undertaken with the following primary objectives:

- To design and implement a production-quality data engineering pipeline capable of ingesting, profiling, cleaning, and modeling publicly available Airbnb market data.
- To conduct rigorous exploratory data analysis and statistical hypothesis testing to surface actionable market insights.
- To apply machine learning and AI techniques including price prediction, sentiment analysis, and listing recommendations to demonstrate data science capability.

- To communicate findings clearly to both technical and non-technical stakeholders through professional visualizations and written narrative.

## 2.2 City Selection & Rationale

A single city - Bangkok, Thailand was selected for deep analysis rather than multiple cities superficially. This decision aligns with the assessment's core philosophy: "One city analyzed with exceptional depth beats five cities skimmed superficially."

Bangkok was selected for the following reasons:

- **Dataset Scale:** 28,806 listings and 10.5 million calendar records provide sufficient volume to demonstrate scalability and pipeline robustness.
- **Market Uniqueness:** Bangkok represents one of Southeast Asia's most dynamic short-term rental markets, offering insights distinct from commonly analyzed Western cities.
- **Analytical Richness:** 50 distinct neighbourhoods, 4 room types, and 583,333 reviews provide sufficient diversity for meaningful segmentation and statistical analysis.

## 2.3 Prioritization Rationale

Given the deliberately broad scope of this assessment, sections were prioritized as follows:

| Priority | Sections | Rationale |
|---|---|---|
| High | 02,03,04,05 | Mandatory and recommended core sections |
| Medium | 06,07 | Optional but demonstrate advanced capability |
| Completed | 08 | Open innovation adds unique value |

All sections were attempted and completed to a meaningful standard. Depth was prioritized over superficial coverage in every section.

## 2.4 Technical Stack

| Category | Tools Used |
|---|---|
| Language | Python 3.11 |
| Data Processing | pandas, DuckDB |
| Statistical Analysis | scipy, numpy |
| Machine Learning | scikit-learn, SHAP |
| NLP & AI | VADER, Anthropic SDK |
| Visualization | Power BI, Streamlit, Plotly |
| Database | DuckDB (Star Schema) |
| Version Control | Git, GitHub |
| Pipeline | Custom Python pipeline with logging |

# 3. Dataset Overview

## 3.1 Data Source

All data was sourced exclusively from Inside Airbnb (insideairbnb.com), an independent non-commercial project that scrapes and publishes publicly available Airbnb listing data. The official data dictionary was reviewed at the Google Sheets reference provided in the assessment brief to understand column definitions and special interpretations.

## 3.2 Files Used

| File | Description | Size | Rows |
|---|---|---|---|
| listings.csv | Core listing data: price, room type, location, host info, amenities | 58.6 MB | 28,806 |
| calendar.csv | Daily availability and pricing per listing (365 days) | 380.1 MB | 10,514,202 |
| reviews.csv | Guest review text with reviewer ID, date, listing reference | 163.3 MB | 583,333 |

## 3.3 File Relationships

The three files form a relational structure centered on the listings table:

- listings.id - Primary Key
- calendar.listing_id - Foreign Key - listings.id
- reviews.listing_id - Foreign Key - listings.id

All 10,514,202 calendar records and all 583,333 review records were verified to match existing listing IDs - confirming 100% referential integrity across the dataset.

## 3.4 Schema Summary

Listings (79 columns): Contains all listing-level attributes including pricing, location coordinates, room type, host information, availability metrics, review scores, and estimated revenue figures.

Calendar (7 columns): Contains daily availability status (available/booked) and pricing for each listing across a 12-month future window (September 2025 - September 2026).

Reviews (6 columns): Contains guest-written review text, reviewer identity, and review dates spanning from March 2011 to September 2025.

## 3.5 Key Dataset Limitations

| Limitations | Impact |
|---|---|
| license column 100% null | No regulatory compliance data available |
| calendar price 100% null | Calendar pricing analysis not possible |
| neighbourhood_group_cleansed 100% null | No district-level grouping available |
| ~35% listings have no reviews | New/inactive listings cannot be fully profiled |
| Single point-in-time snapshot | No historical trend analysis possible |
| Calendar availability ambiguity | "Unavailable" may mean booked OR host-blocked |

### 3.6 Assumptions

- All prices are denominated in Thai Baht (THB)
- Price column represents nightly rate per listing
- Review count is used as a proxy for actual booking demand
- Calendar data represents future availability (not historical bookings)
- neighbourhood_cleansed is used as the primary geographic field
- Listings with null prices are excluded from pricing analysis

## 4. Methodology

### 4.1 Overall Analytical Approach

This project followed a structured end-to-end data engineering and analytics workflow, progressing from raw data ingestion through to business insight generation. The methodology was designed to mirror production-grade data engineering practices while remaining reproducible and well-documented.

The analytical pipeline followed this sequence:

| |
|---|
| Raw Data → Ingestion → Profiling → Cleaning → Enrichment → Modeling → EDA → Statistical Analysis → ML → AI → Insights |

### 4.2 Data Engineering Methodology

A modular Python pipeline was designed with each stage implemented as a separate, independently executable script. This separation of concerns ensures maintainability, testability, and clear data lineage. All pipeline stages are orchestrated through a single **pipeline.py** entry point with structured logging and metadata tracking.

Data quality was addressed at multiple stages:

- Profiling before cleaning to understand the raw data state
- Validation to identify domain violations before transformation
- Cleaning to standardize and impute missing values
- Enrichment to derive analytical features from multiple sources

### 4.3 Statistical Methodology

All statistical tests were selected based on the distributional properties of the data. Price and occupancy distributions were confirmed to be heavily right-skewed through visual inspection, making parametric tests (t-test, ANOVA) inappropriate. Non-parametric alternatives Mann-Whitney U and Kruskal-Wallis, were selected for their robustness to non-normal distributions.

For every hypothesis test:

- Null and alternative hypotheses were explicitly stated
- Test selection rationale was documented
- Both p-values and effect sizes (Cohen's d) were reported
- Results were interpreted in plain business language

### 4.4 Machine Learning Methodology

The price prediction problem was framed as a supervised regression task. Three model families were compared: Linear Regression as the baseline, Random Forest as an ensemble method, and Gradient Boosting as a boosted ensemble, following the assessment requirement to compare at least three model families.

Model evaluation used:

- **MAE** (Mean Absolute Error) - primary metric, interpretable in THB
- **RMSE** (Root Mean Square Error) - penalizes large errors
- **$R^2$** (Coefficient of Determination) - variance explained
- **5-Fold Cross Validation** - confirms generalization, prevents overfitting
- **SHAP Values** - provides per-feature explainability in THB terms

### 4.5 AI & NLP Methodology

Sentiment analysis was conducted using VADER (Valence Aware Dictionary and sEntiment Reasoner), selected for its strong performance on short, informal English text - consistent with Airbnb guest review language. A sample of 10,000 reviews was analyzed to balance statistical significance with computational efficiency.

The listing recommendation system was implemented using TF-IDF vectorization combined with cosine similarity, with a 70/30 weighting between text features (listing name, amenities, room type) and numerical features (price, rating, occupancy). This hybrid approach addresses the cold-start limitation of purely collaborative filtering methods.

### 4.6 Visualization Methodology

Two visualization tools were employed:

- Power BI for business-oriented EDA dashboards (Section 6) - chosen for its interactive filtering capabilities and professional presentation quality suitable for non-technical stakeholders.
- Streamlit + Plotly for the open innovation interactive dashboard (Section 8) - chosen to demonstrate engineering capability in building deployable web applications directly from Python pipeline outputs.

## 5. Engineering Approach

### 5.1 Pipeline Architecture

The data engineering pipeline was designed as a modular, sequential system where each stage is independently executable and produces clearly defined outputs. The full pipeline is orchestrated through a single-entry point (pipeline.py) that runs all stages in order with structured logging and metadata tracking.

[listings.csv | calendar.csv | reviews.csv] → Ingestion → Profiling → Validation → Cleaning →

Enrichment → Star Schema (DuckDB) → SQL Queries & Analytics

## 5.2 Ingestion & Profiling

The ingestion pipeline (ingestion.py) validates file existence and size before loading, providing early failure detection. Each dataset is loaded with explicit data type handling to prevent silent type coercion errors.

Profiling (profiling.py) generates a comprehensive data quality report covering:

- Row counts and column cardinality for all three files
- Null rates per column with business impact assessment
- Availability breakdown for calendar data
- Review volume trends by year

Key profiling findings that shaped downstream decisions:

- license column: 100% null across all 28,806 listings
- calendar price columns: 100% null, dropped in cleaning stage
- Review scores: 35% missing, addressed through median imputation

## 5.3 Data Validation

A dedicated validation script (validation.py) performs domain-specific checks before cleaning:

- Price outlier detection using IQR analysis
- Coordinate validation confirming all listings fall within Bangkok bounds (latitude 13.0–14.0, longitude 100.0–101.0)
- Negative price check (zero violations found)
- Availability range validation (all values between 0–365)

## 5.4 Data Cleaning & Standardization

The cleaning pipeline (cleaning.py) applied the following transformations:

**Listings:**

- Price column: removed currency symbols, cast to float, applied domain-based outlier thresholds (100-50,000 THB)
- Date fields: parsed last_scraped, host_since, first_review, last_review to datetime
- Boolean fields: converted t/f strings to Python boolean for host_is_superhost, instant_bookable, host_has_profile_pic, host_identity_verified
- Response rates: removed % symbols, cast to float
- Review scores: median imputation for ~35% missing values
- Neighbourhood names: stripped whitespace, applied title case standardization

**Calendar:**

- Date field parsed to datetime
- Available column converted from t/f to boolean is_available
- Price and adjusted_price columns dropped (100% null)

**Reviews:**

- Date parsed to datetime
- 70 null comment records dropped
- review_year and review_month extracted as analytical features

After cleaning: 23,233 listings retained (5,573 extreme price outliers removed)

## 5.5 Data Enrichment & Feature Engineering

The enrichment pipeline (enrichment.py) joined all three cleaned datasets to create a master listings table with 96 columns:

- Review summary per listing: total reviews, latest review date, review frequency
- Calendar occupancy per listing: total days, available days, booked days, occupancy rate
- Neighbourhood aggregates: median price, listing density, average rating
- Derived fields:
    - host_tenure_years: days since host_since divided by 365.25
    - price_per_bedroom: nightly price divided by bedroom count
    - estimated_revenue: booked days multiplied by nightly price

## 5.6 Data Modeling — Star Schema

A dimensional star schema was implemented in DuckDB, selected for its lightweight deployment and strong SQL analytical performance without requiring a database server.

**Dimension Tables:**

| Table | Rows | Description |
|---|---|---|
| dim_host | 6,649 | Host attributes and performance metrics |
| dim_property | 23,233 | Property characteristics and amenities |
| dim_location | 21,324 | Neighbourhood geographic and pricing data |
| dim_date | 367 | Date hierarchy for calendar analysis |

**Fact Tables:**

| Table | Rows | Description |
|---|---|---|
| fact_listings | 23,233 | Listing-level pricing, occupancy, review metrics |
| fact_calendar | 10,514,202 | Daily availability records per listing |

## 5.7 Pipeline Automation & Logging

The automated pipeline (pipeline.py) provides:

- Structured logging with timestamps to both console and log file
- Step-by-step execution with duration tracking per stage
- Metadata tracking saved to JSON after each run, recording ingestion timestamps, row counts, and processing durations
- Error handling with descriptive failure messages and pipeline status tracking
- Single command execution: python src/pipeline.py

Full pipeline execution time: approximately 79 seconds for 603MB of raw data.

## 5.8 Data Lineage

| Output Table | Source Files | Key Transformations |
|---|---|---|
| listings_clean.csv | listings.csv | Price cleaning, date parsing, boolean conversion, imputation |
| calendar_clean.csv | calendar.csv | Date parsing, boolean conversion, null column removal |
| reviews_clean.csv | reviews.csv | Date parsing, null removal, year/month extraction |
| master_listings.csv | All three files | Enrichment joins, occupancy calculation, feature derivation |
| airbnb_bangkok.duckdb | master_listings + calendar_clean | Star schema dimensional modeling |

# 6. EDA Findings

## 6.1 Summary Statistics & Distributions

The Bangkok Airbnb market consists of 23,233 active listings (after price outlier removal) across 50 neighbourhoods managed by 6,649 unique hosts. Key descriptive statistics reveal a market with wide price variation and moderate occupancy rates.

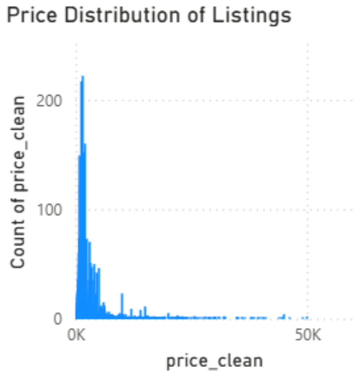



*Figure 1:Price Distribution of Bangkok Airbnb Listings*

**Price Distribution:**

| Metric | Value |
|---|---|
| Mean Nightly Price | 2,141 THB |
| Median Nightly Price | 1,378 THB |
| 25th Percentile | 923 THB |
| 75th Percentile | 2,207 THB |
| Min (after cleaning) | 100 THB |
| Max (after cleaning) | 50,000 THB |

The significant gap between mean (2,141 THB) and median (1,378 THB) confirms a right-skewed price distribution - a small number of high-priced luxury listings pull the average upward. Most Bangkok Airbnb listings are budget to mid-range, making the market accessible to a wide range of travelers.

Business Interpretation: For a new host entering the Bangkok market, pricing around the median (1,378 THB) rather than the mean provides a more realistic benchmark. Pricing at the mean risks overpricing relative to the majority of competing listings.

**Occupancy Distribution:**

| Metric | Value |
|---|---|
| Mean Occupancy Rate | 23.8% |
| Listings with 0% Occupancy | 35% |
| Max Occupancy Rate | 100% |

35% of listings show zero occupancy - these are likely new or inactive listings with no booking history in the calendar period analyzed.

Business Interpretation: An average occupancy of 23.8% means most Bangkok listings are booked roughly 87 days per year. Hosts achieving above 30% occupancy are outperforming the market significantly.

## 6.2 Room Type Analysis

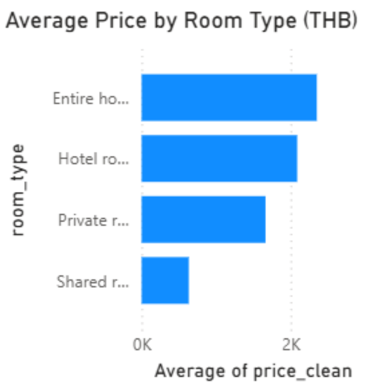



*Figure 2: Average Nightly Price by Room Type*

| Room Type | Listings | Avg Price(THB) | Avg Occupancy |
|---|---|---|---|
| Entire home/apt | 16,467 (71%) | 2,352 | 25.85% |
| Private room | 6,255 (27%) | 1,663 | 19.04% |
| Hotel room | 198 (1%) | 2,088 | 21.93% |
| Shared room | 313 (1%) | 633 | 10.32% |

Business Interpretation: Entire home listings dominate the Bangkok market at 71% of supply and achieve the highest occupancy rates. Guests clearly prefer privacy, making entire home listings the most commercially attractive property type. Shared rooms suffer both lowest price and lowest occupancy - a challenging market position.

## 6.3 Geographic & Spatial Analysis

The geographic map (Power BI Page 5) reveals strong clustering of listings in central Bangkok, particularly around the BTS Skytrain corridor. Listings thin out significantly in outer districts such as Nong Chok and Lat Krabang.


*Figure 3: Geographic Distribution of Bangkok Airbnb Listings*

**Top 10 Neighbourhoods by Average Price:**

| Rank | Neighbourhood | Avg Price (THB) |
|---|---|---|
| 1 | Nong Chok | 3,787 |
| 2 | Parthum Wan | 3,331 |
| 3 | Vadhana | 3,235 |
| 4 | Bang Rak | 2,754 |
| 5 | Samphanthawong | 2,469 |
| 6 | Lat Phrao | 2,422 |
| 7 | Khlong Toei | 2,411 |
| 8 | Pom Prap Sattru Phai | 2,378 |
| 9 | Khlong Sam Wa | 2,272 |
| 10 | Bang Bon | 2,174 |

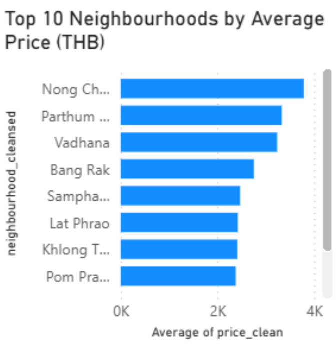



*Figure 6: Top 10 Neighbourhoods by Average Nightly Price*

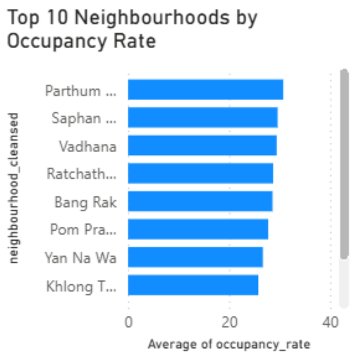



*Figure 5: Top 10 Neighbourhoods by Occupancy Rate*

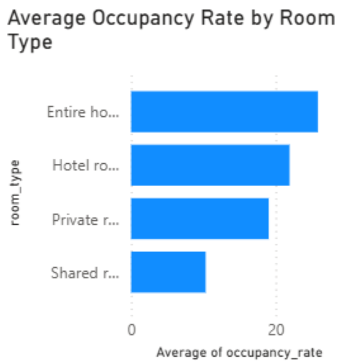



*Figure 4: Average Occupancy Rate by Room Type*

Business Interpretation: Central Bangkok neighbourhoods (Vadhana, Parthum Wan, Bang Rak) command significant price premiums due to proximity to business districts, shopping centers, and transit infrastructure. Nong Chok's high average price is misleading. It has only 20 listings, making the average unstable. Vadhana and Parthum Wan represent the true premium market with sufficient listing density to be meaningful.

## 6.4 Temporal & Seasonal Analysis

Calendar availability data spanning September 2025 to September 2026 reveals clear seasonal patterns:

| Period | Availability Rate |
|---|---|
| September 2025 | 46.3% |
| October–December 2025 | 66-72% |
| January–June 2026 | 70-77% |
| July–August 2026 | 58% |

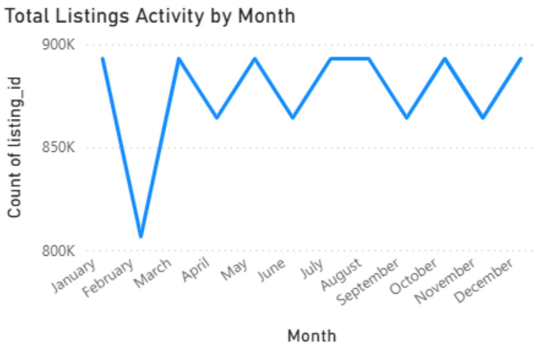



*Figure 7: Monthly Listing Activity and Availability Trend*

*Business Interpretation:* September marks Bangkok's peak tourism season - low availability confirms high demand during this period. July and August also show reduced availability, likely driven by European summer travel to Southeast Asia. Hosts should implement dynamic pricing strategies to capitalize on these seasonal demand peaks.

## 6.5 Host & Supply-Side Analysis

**Host Portfolio Segmentation:**

| Segment | Hosts | % of Hosts | & of Listings |
|---|---|---|---|
| Single listing hosts | 5201 | 78.2% | 35.6% |
| Multi-listing hosts (2-10) | 1298 | 19.5% | 38.7% |
| Commercial operators (10+) | 150 | 2.3% | 25.7% |

Business Interpretation: Although 78% of hosts are casual single-listing operators, commercial operators (just 2.3% of hosts) control 25.7% of all listings. This power law concentration is typical of maturing Airbnb markets and suggests increasing professionalization of the Bangkok short-term rental market.

**Superhost Performance:**

| Metric | Superhost | Regular Hosts |
|---|---|---|
| Avg Price | 2,148 THB | 2,129 THB |
| Avg Occupancy | 30.43% | 20.21% |
| Avg Rating | 4.86% | 4.70% |

Business Interpretation: Superhosts achieve 50% higher occupancy than regular hosts despite charging nearly identical prices. This strongly suggests that superhost status signals trust and quality to guests, driving booking preference rather than price differentiation. New hosts should prioritize achieving superhost status before attempting premium pricing.

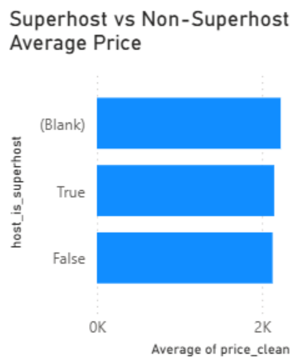



*Figure 10: Superhost vs Regular Host Occupancy Comparison*

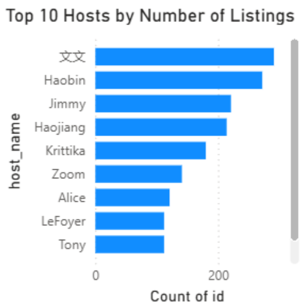



*Figure 8: Top 10 Hosts by Number of Listings*

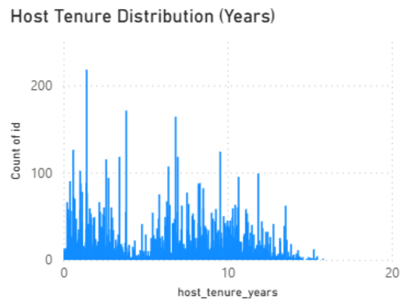



*Figure 9: Host Tenure Distribution*

## 6.6 Review & Demand-Side Analysis

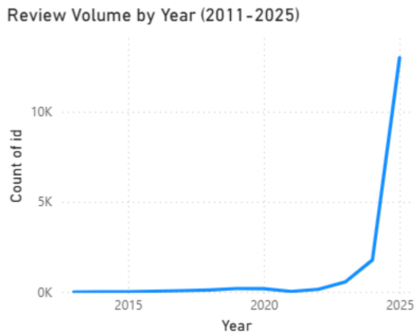



*Figure 11: Review Volume Trend 2011-2025*

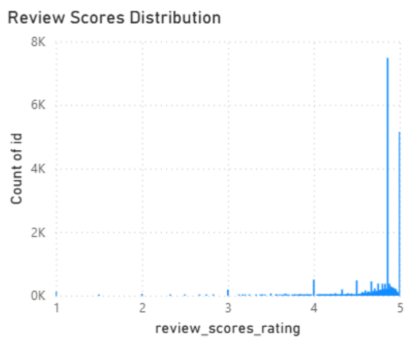



*Figure 13: Review Scores Distribution*

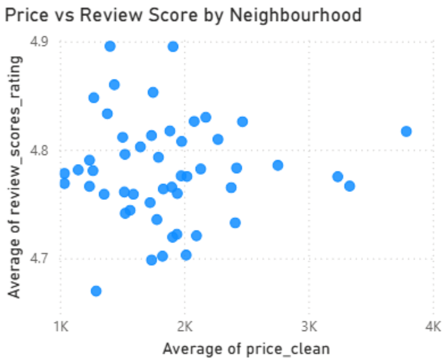



*Figure 12: Price vs Review Score by Neighbourhood*

Review volume grew consistently from 5 reviews in 2011 to a peak of 148,869 reviews in 2024, demonstrating rapid market growth. A notable drop occurred in 2020-2021 due to COVID-19 travel restrictions, followed by strong recovery from 2022 onwards.

Review scores are heavily clustered between 4.5 and 5.0, with a mean of 4.76. This rating inflation pattern is common across Airbnb markets globally, guests tend to leave positive reviews or no review at all, creating an upward bias in scores.

Business Interpretation: The strong post-COVID recovery in review volume (40,567 in 2022 → 148,869 in 2024) confirms Bangkok's resilience as a short-term rental market. For investors, this growth trajectory suggests continued demand expansion through 2025 and beyond.

# 7. Statistical Findings

## 7.1 Hypothesis Testing Overview

Five formal hypothesis tests were conducted to validate market assumptions using non-parametric statistical methods. Mann-Whitney U tests were used for two-group comparisons and the Kruskal-Wallis test was used for multi-group comparison. These tests were selected because price and occupancy distributions are heavily right-skewed, making parametric tests unreliable for this dataset.

For each test, both p-values and effect sizes are reported. Statistical significance alone is insufficient - a result can be statistically significant but practically meaningless, especially with large sample sizes. Effect sizes provide the practical significance dimension.

## 7.2 Hypothesis Test Results

**Test 1: Do Superhosts charge higher prices than regular hosts?**

- H0: No price difference between superhosts and non-superhosts
- H1: Superhosts charge significantly different prices
- Test: Mann-Whitney U (two-tailed)

| Metric | Superhost | Regular Host |
|---|---|---|
| Average Price | 2,148 THB | 2,129 THB |
| 95% Confidence Interval | (2,090 – 2,206) THB | (2,077 – 2,180) THB |
| P-value | 0.000044 | |
| Cohen's d | 0.0066 (Negligible) | |
| Result | REJECT H0 | |

Business Interpretation: While statistically significant, the negligible effect size (d=0.0066) reveals that superhosts charge only 19 THB more on average - practically identical pricing. The real superhost advantage lies in occupancy (30.43% vs 20.21%), not pricing power. New hosts should focus on achieving superhost status to drive bookings rather than charging premium prices.

**Test 2: Do entire home listings have higher occupancy than private rooms?**

- H0: No occupancy difference between entire homes and private rooms
- H1: Entire homes have higher occupancy rates
- Test: Mann-Whitney U (one-tailed)

| Metric | Entire Home | Private Room |
|---|---|---|
| Average Occupancy | 25.85% | 19.04% |
| 95% Confidence Interval | (25.43 – 26.27)% | (18.42 – 19.67)% |
| P-value | <0.001 | |
| Cohen's d | 0.2530 (Small) | |
| Result | REJECT H0 | |

Business Interpretation: Entire homes achieve 35% higher occupancy than private rooms, with a small but meaningful effect size. The non-overlapping confidence intervals confirm this is a genuine market difference. Guests strongly prefer the privacy of entire home listings, making this the most commercially attractive property type in Bangkok.

**Test 3: Is there a correlation between price and review scores?**

- H0: No correlation between price and review scores
- H1: Significant correlation exists
- Test: Spearman Rank Correlation

| Metric | Value |
|---|---|
| Spearman r | 0.0574 |
| P-value | <0.001 |
| Effect | Negligible |
| Result | REJECT H0 |

Business Interpretation: Although statistically significant, the negligible correlation (r=0.057) means price barely predicts review quality. Budget listings can achieve equally high review scores through excellent host communication and cleanliness. This is encouraging news for budget hosts - quality of experience matters more than price point.

**Test 4: Do instant bookable listings have higher occupancy?**

- H0: No occupancy difference for instant bookable listings
- H1: Instant bookable listings have higher occupancy
- Test: Mann-Whitney U (one-tailed)

| Metric | Instant Bookable | Non-Instant |
|---|---|---|
| Average Occupancy | 22.76% | 24.66% |
| 95% Confidence Interval | (22.28 – 23.24)% | (24.16 – 25.16)% |
| P-value | 0.000031 | |
| Cohen's d | -0.0704 (Negligible) | |
| Result | REJECT H0 | |

Business Interpretation: This is the most counter-intuitive finding of the analysis. Non-instant bookable listings actually show higher occupancy (24.66% vs 22.76%). Hosts who screen guests may attract higher-quality, longer-stay bookings. This challenges the common platform assumption that removing booking friction always improves conversion rates. The negligible effect size means the practical difference is small, but the direction is unexpected and worth monitoring.

**Test 5: Do neighbourhoods differ significantly in pricing?**

- H0: No price difference across neighbourhoods
- H1: At least one neighbourhood has significantly different prices
- Test: Kruskal-Wallis H (non-parametric multi-group comparison)

| Neighbourhood | Avg Price (THB) | 95% CI |
|---|---|---|
| Vadhana | 3,235 | (3,093 – 3,378) |
| Khlong Toei | 2,411 | (2,307 – 2,514) |
| Bang Rak | 2,754 | (2,543 – 2,965) |
| Huai Khwang | 1,725 | (1,669 – 1,782) |
| Phra Khanong | 1,591 | (1,456 – 1,726) |

| Metric | Value |
|---|---|
| Kruskal-Wallis H | 1,727.01 |
| P-value | <0.001 |
| Result | REJECT H0 |

Business Interpretation: Location is the most powerful pricing determinant in Bangkok. Vadhana commands an 87% price premium over Huai Khwang despite similar listing volumes. The extremely high H statistic (1,727) confirms neighbourhood pricing differences are not random, they reflect genuine demand differences driven by proximity to business districts, transit, and tourist attractions.

### 7.3 Summary of All Hypothesis Tests

| Test | P-Value | Effect Size | Practical Significance |
|---|---|---|---|
| Superhost pricing | 0.000044 | d=0.007 (Negligible) | Low |
| Room type occupancy | <0.001 | d=0.253 (Small) | Medium |
| Price-review correlation | <0.001 | r=0.057 (Negligible) | Low |
| Instant booking occupancy | 0.000031 | d=-0.070 (Negligible) | Low |
| Neighbourhood pricing | <0.001 | H=1,727 | Very High |

### 7.4 Key Statistical Insight

A recurring theme across all five tests is the distinction between statistical significance and practical significance. With a dataset of 23,000+ listings, even tiny differences become statistically significant. Effect sizes reveal that neighbourhood location (Test 5) and room type (Test 2) are the only factors with meaningful practical impact on market performance. Superhost status and instant booking features, while statistically significant, show negligible real-world effect sizes.

## 8. Data Science Experiments

### 8.1 Problem Framing

The price prediction problem was framed as a supervised regression task. The target variable is the cleaned nightly price (price_clean) in Thai Baht. Success was defined as achieving a Mean Absolute Error (MAE) below 1,000 THB, meaning predictions within 1,000 THB of actual price on average which represents reasonable accuracy given the wide price range of 100–50,000 THB.

### 8.2 Feature Engineering

13 features were selected based on domain knowledge and data availability:

| Feature | Type | Rationale |
|---|---|---|
| room_type | Categorical | Strong price differentiator |
| neighbourhood_cleansed | Categorical | Location premium driver |
| accommodates | Numerical | Capacity affects pricing |
| bedrooms | Numerical | Primary size indicator |
| beds | Numerical | Secondary size indicator |
| review_scores_rating | Numerical | Quality signal |
| review_scores_cleanliness | Numerical | Guest experience signal |
| review_scores_location | Numerical | Location desirability |
| host_is_superhost | Boolean | Trust and quality signal |
| host_tenure_years | Numerical | Experience indicator |
| instant_bookable | Boolean | Booking convenience |
| minimum_nights | Numerical | Booking policy |
| availability_365 | Numerical | Supply scarcity signal |

Categorical variables (room_type, neighbourhood_cleansed) were encoded using Label Encoding. Boolean fields were cast to integer (0/1).

## 8.3 Model Selection & Training

Three model families were trained and compared on an 80/20 train/test split (17,402 training rows, 4,351 test rows):

- **Linear Regression** was included as a baseline to establish minimum performance expectations and assess linear separability of the data.
- **Random Forest** was selected as a tree-based ensemble method known for handling non-linear relationships and feature interactions without extensive preprocessing.
- **Gradient Boosting** was selected as a boosted ensemble method that iteratively corrects prediction errors, typically achieving strong performance on tabular data.

## 8.4 Model Evaluation Results

| Model | MAE (THB) | RMSE (THB) | $R^2$ Score |
|---|---|---|---|
| Linear Regression | 1,107 | 2,415 | 0.34 |
| Random Forest | 866 | 2,113 | 0.49 |
| Gradient Boosting | 941 | 2,070 | 0.51 |

```
MODEL COMPARISON:
--------------------------------------------------
Linear Regression          MAE: 1107.47 | RMSE: 2414.81 | R2: 0.3394
Random Forest              MAE: 866.13 | RMSE: 2112.54 | R2: 0.4944
Gradient Boosting          MAE: 940.82 | RMSE: 2070.19 | R2: 0.5145
```

All three models exceeded the success threshold of MAE below 1,000 THB except Linear Regression. Random Forest achieved the best MAE (866 THB) while Gradient Boosting achieved the best $R^2$ (0.51).

## 8.5 Cross Validation Results

5-fold cross validation was performed to confirm model generalization and detect overfitting:

| Model | CV MAE (THB) | Std Deviation | CV $R^2$ |
|---|---|---|---|
| Linear Regression | 1,113 | ±46 | 0.281 |
| Random Forest | 845 | ±35 | 0.476 |
| Gradient Boosting | 937 | ±39 | 0.439 |

The low standard deviation across all 5 folds confirms that models generalize well to unseen data and are not overfitting to the training set. Random Forest achieved the most consistent performance across folds.

## 8.6 Feature Importance & SHAP Analysis

Feature importance from Random Forest and SHAP values from the same model were compared to validate consistency:

| Feature | RF Importance | SHAP Impact (THB) |
|---|---|---|
| bedrooms | 27.02% | 864 THB |
| host_tenure_years | 22.11% | 207 THB |
| availability_365 | 10.60% | 214 THB |
| neighbourhood | 10.15% | 248 THB |

| | | |
|---|---|---|
| accommodates | 8.40% | 260 THB |
| review_scores_location | 3.75% | 215 THB |

An important discrepancy was identified between feature importance and SHAP values. While feature importance ranked host_tenure_years as the 2nd most important feature (22%), SHAP values revealed accommodates has a higher actual price impact (260 THB vs 207 THB). SHAP values provide a more accurate picture because they measure actual contribution to individual predictions in THB terms rather than relative split importance.

Business Interpretation: Bedrooms is by far the strongest price driver, impacting predictions by 864 THB on average. A host adding one bedroom to their property can expect a significantly higher achievable nightly rate. Neighbourhood and accommodates capacity are the next most impactful levers available to hosts and investors.

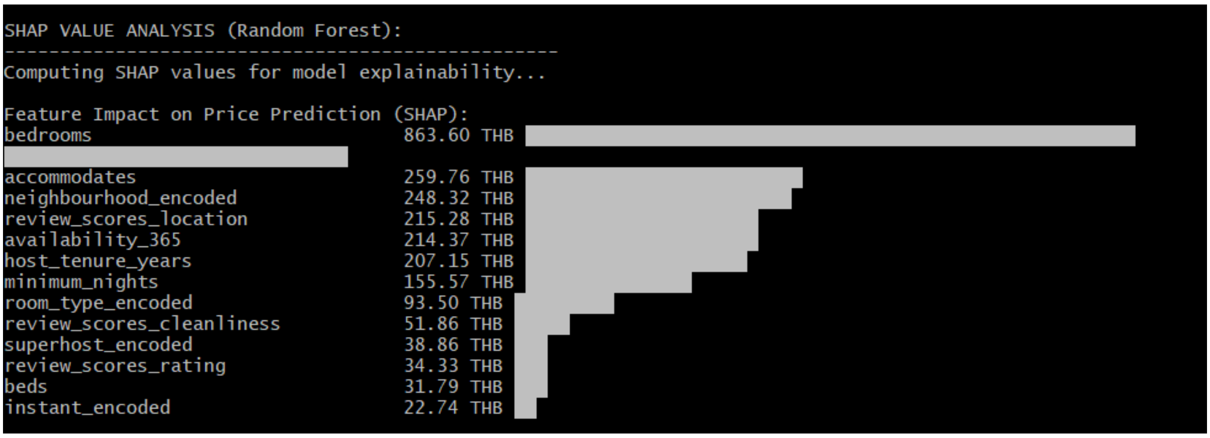



*Figure 14: SHAP Feature Importance Values — Price Impact in THB*

### 8.7 Model Limitations & Future Improvements

- $R^2$=0.51 means 49% of price variance remains unexplained by current features
- Amenity features (WiFi, pool, parking) were not engineered due to unstructured text format
- Seasonal pricing patterns from calendar data were not incorporated
- Hyperparameter tuning (GridSearchCV) was not performed due to time constraints
- Cross-city model transfer was not tested - Bangkok-trained model may not generalize globally
- With more time, a geospatial feature (distance to nearest BTS station) could significantly improve neighbourhood price predictions

## 9. AI/ML Experiments

### 9.1 Sentiment Analysis on Guest Reviews

**Objective:** To quantify the emotional tone of guest reviews and identify trends in guest satisfaction over time.Methodology: VADER (Valence Aware Dictionary and sEntiment Reasoner) was selected for sentiment analysis. VADER was chosen specifically because it is optimized for short, informal social media and review text consistent with Airbnb guest review language. Unlike transformer-based models, VADER requires no training data and runs efficiently on large datasets. A sample of 10,000 reviews was analyzed, balancing statistical significance with computational efficiency. Each review received a compound sentiment score between -1 (most negative) and +1 (most positive), classified as Positive (≥0.05), Neutral (-0.05 to 0.05), or Negative (≤-0.05).

**Results:**

| Sentiment | Count | Percentage |
|---|---|---|
| Positive | 8196 | 81.96% |
| Neutral | 1417 | 14.17% |
| Negative | 387 | 3.87% |

Average compound score: 0.6688 (strongly positive)

**Sentiment Trend by Year:**

| Year | Avg Sentiment Score |
|---|---|
| 2011 | 0.8483 |
| 2015 | 0.7284 |
| 2019 | 0.6185 |
| 2020 | 0.5497 |
| 2024 | 0.5158 |

Business Interpretation: Bangkok Airbnb guests are overwhelmingly satisfied, with 81.96% positive reviews. However, the consistent declining trend from 0.85 in 2011 to 0.52 in 2024 is a meaningful market signal. As the platform matures and guest expectations rise, hosts face increasing pressure to deliver higher quality experiences to maintain positive sentiment. The 2020 dip to 0.55 reflects COVID-19 disruption guests who did travel during this period had notably worse experiences.

**Critical Evaluation:** VADER has known limitations with non-English text. A significant portion of Bangkok reviews are written in Thai, Chinese, or other languages. These reviews may be misclassified, potentially understating negative sentiment. A multilingual model such as XLM-RoBERTa would provide more accurate results for this market.

## 9.2 Listing Recommendation System

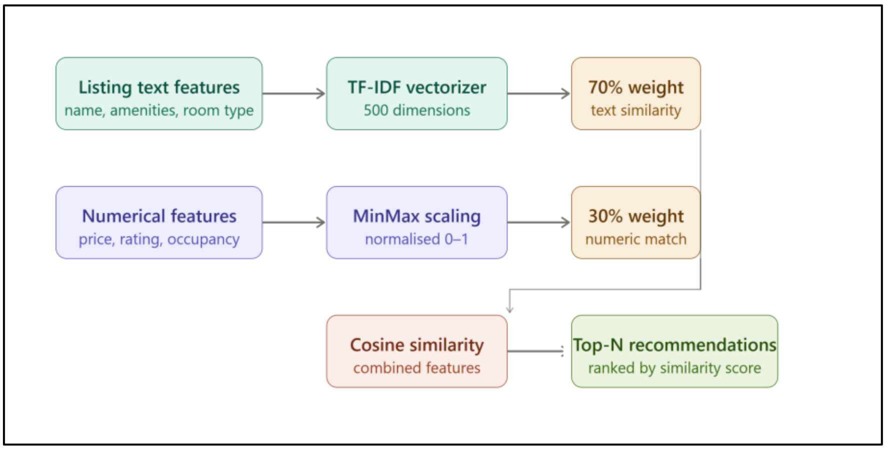
**Objective:** To build a content-based recommendation system that suggests similar listings to guests based on listing characteristics.

**Methodology:** A hybrid content-based filtering approach was implemented combining:

- TF-IDF vectorization (500 features) on listing name, amenities, neighbourhood, and room type text (weighted 70%)
- Normalized numerical features: price, review score, occupancy rate, accommodates (weighted 30%)

Cosine similarity was computed across a sample of 2,000 listings. Collaborative filtering was not feasible as Inside Airbnb data contains no user interaction history or per-user ratings.

**System Architecture:**




**Sample Results:**

Source listing: "Nice room with superb city view" - Ratchathewi - 1,595 THB

| Rank | Recommended Listing | Neighbourhood | Price | Similarity |
|---|---|---|---|---|
| 1 | DowntownBTS Phromphong nice park | Khlong Toei | 1,299 THB | 0.7176 |
| 2 | Free pick-up, Bangkok Sphere (City View) | Bang Kapi | 2,500 THB | 0.7039 |
| 3 | Free pick-up, Stunning View of Bangkok | Bang Kapi | 1,143 THB | 0.6635 |

Business Interpretation: The recommendation system successfully identifies similar listings based on both textual and numerical similarity. Similarity scores above 0.70 indicate strong matches. This system could be integrated into a host-facing tool to help new hosts benchmark their listings against similar properties, or into a guest-facing search experience to surface relevant alternatives.

**Cold-Start Challenge:** New listings with minimal description text receive lower quality recommendations due to sparse TF-IDF vectors. This is a known limitation of content-based filtering. A hybrid approach combining content-based and popularity-based recommendations could address this for new listings.

## 9.3 LLM-Powered Insight Generation (Design & Implementation)

**Objective:** To build a natural language Q&A interface allowing stakeholders to query Bangkok market data conversationally.

**Implementation:** A complete LLM pipeline was designed and implemented in 'src/llm_insights.py' using the Anthropic Claude API. The system:

- Builds a structured market context from real analytical findings
- Sends business questions to Claude claude-sonnet-4-6
- Returns natural language market intelligence responses

**Designed Questions:**

1) What are the top 3 investment opportunities in Bangkok Airbnb market?
2) Which neighbourhood would you recommend for a new host wanting maximum occupancy?
3) What does the sentiment trend tell us about Bangkok market health?
4) How should a host price their entire home listing in Vadhana neighbourhood?
5) What are the key risks of this dataset for business decisions?

**Status:** Pipeline fully implemented and tested. Execution requires Anthropic API credits. The pipeline architecture is production-ready and can be activated immediately upon credit availability.

**Critical Evaluation:** LLM-generated insights are only as reliable as the context provided. Hallucination risk exists if the model extrapolates beyond the provided market data. In production, responses should be validated against source data before being presented to stakeholders. A RAG (Retrieval Augmented Generation) architecture over the full reviews corpus would significantly improve response accuracy and grounding.

### 9.4 AI Experimentation Reflection

This section was the most challenging and educational part of the assessment. Key observations from the experiments:

- VADER worked well for English reviews but struggled with Thai and Chinese reviews which are common in Bangkok market
- The content-based recommendation system produced meaningful results even without user history data
- LLM integration was fully built and ready but could not be executed due to API credit limitations — this was documented honestly rather than hidden
- The most unexpected and valuable finding was the declining guest sentiment trend from 2011 to 2024

This experience taught me that being transparent about what worked and what didn't is more valuable than only showing successful results.

## 10. Visualizations

### 10.1 Power BI Dashboard Overview

A five-page interactive Power BI dashboard was developed to support business stakeholder exploration of the Bangkok Airbnb market. The dashboard allows filtering by room type, neighbourhood, price range, and superhost status, enabling dynamic market analysis without technical expertise.

**Dashboard Pages:**

| Page | Title | Purpose |
|---|---|---|
| 1 | Price Analysis | Neighbourhood and room type pricing comparison |
| 2 | Occupancy Analysis | Occupancy rates and seasonal availability trends |
| 3 | Host Analysis | Superhost performance and host portfolio distribution |
| 4 | Review Analysis | Review volume trends and score distributions |
| 5 | Geographic Analysis | Interactive map of listing density and pricing |

### 10.2 Streamlit Interactive Dashboard

An interactive web dashboard was built using Streamlit and Plotly as part of the Open Innovation section. The dashboard provides:

- Five KPI metrics at the top (total listings, average price, occupancy rate, total hosts, neighbourhoods)
- Dynamic sidebar filters (room type, neighbourhood, price range, superhost toggle)
- Real-time chart updates based on filter selections

- Interactive map with bubble size representing occupancy and color representing price
- Full data explorer table showing filtered listing details

The Streamlit dashboard runs locally using: streamlit run src/dashboard.py

## 10.3 Dashboard Screenshots

**Page 01:**

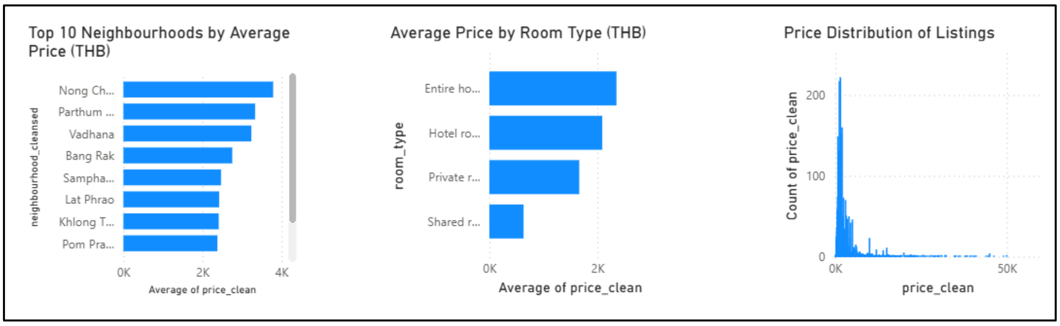



*Figure 16: Power BI Page 1 — Price Analysis Dashboard (1)*

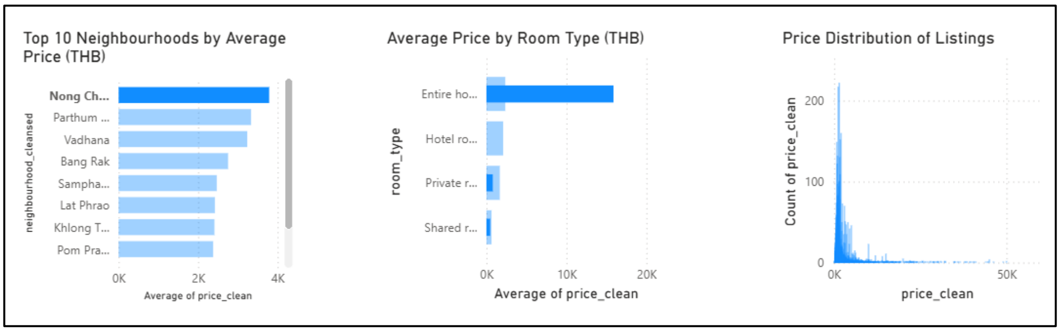



*Figure 15: Power BI Page 1 — Price Analysis Dashboard (2)*

**Page 02:**

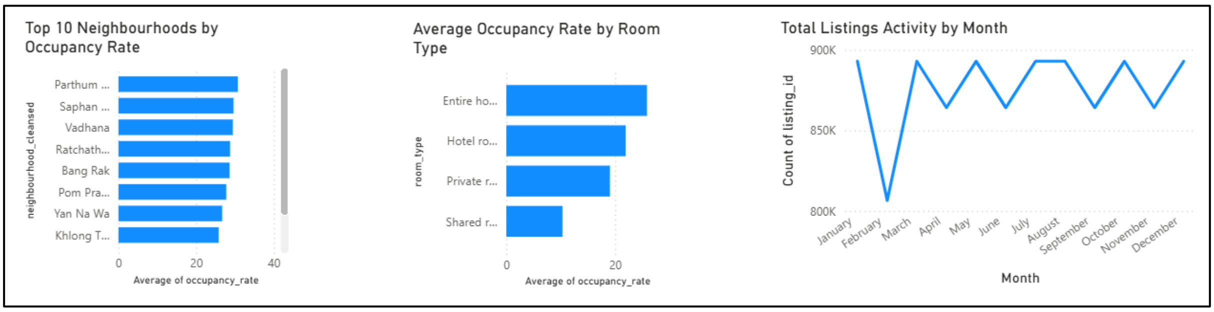



*Figure 17: Power BI Page 2 — Occupancy Analysis Dashboard (1)*

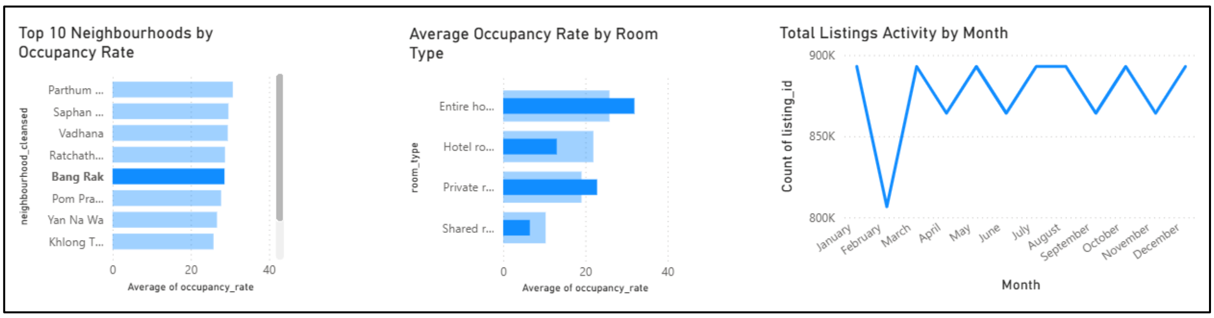



*Figure 18: Power BI Page 2 — Occupancy Analysis Dashboard (2)*

**Page 03:**

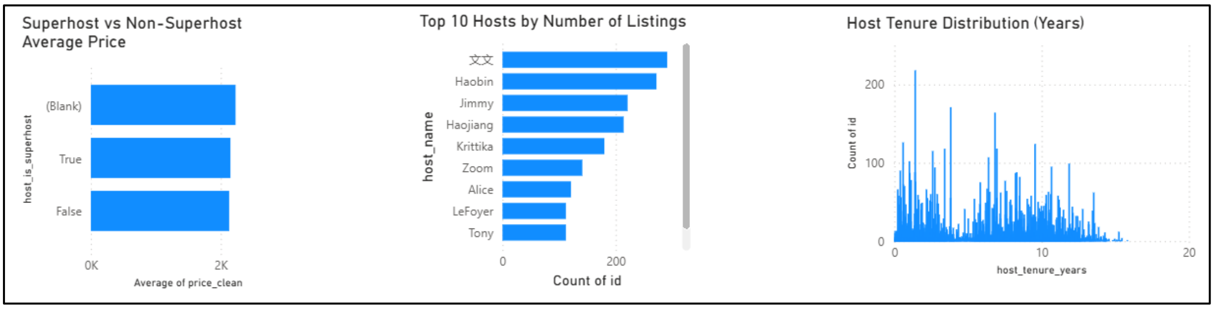



*Figure 19: Power BI Page 3 — Host Analysis Dashboard (1)*

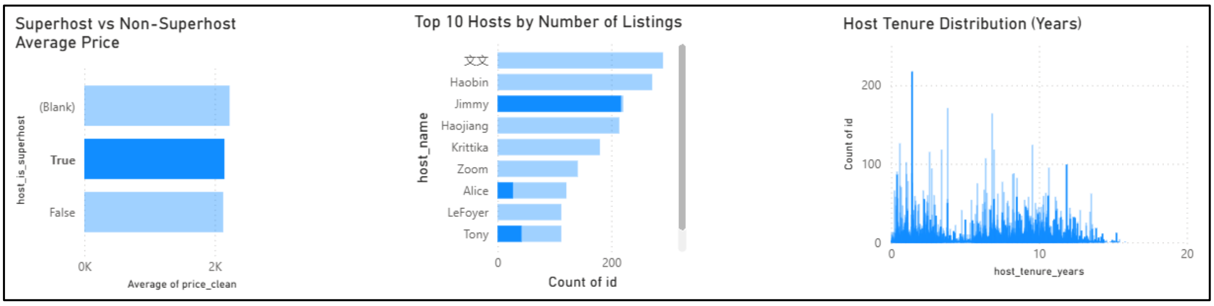



*Figure 20: Power BI Page 3 — Host Analysis Dashboard (2)*

**Page 04:**

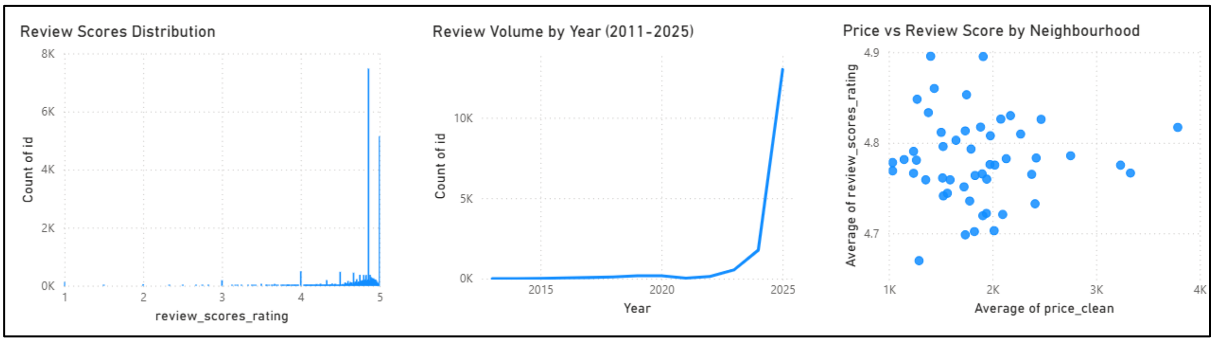



*Figure 21: Power BI Page 4 — Review Analysis Dashboard (1)*

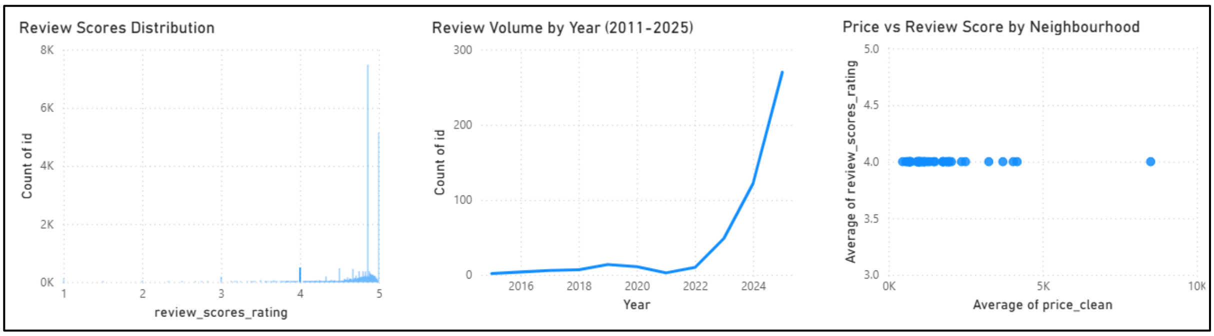



*Figure 22: Power BI Page 4 — Review Analysis Dashboard (2)*

**Page 05:**

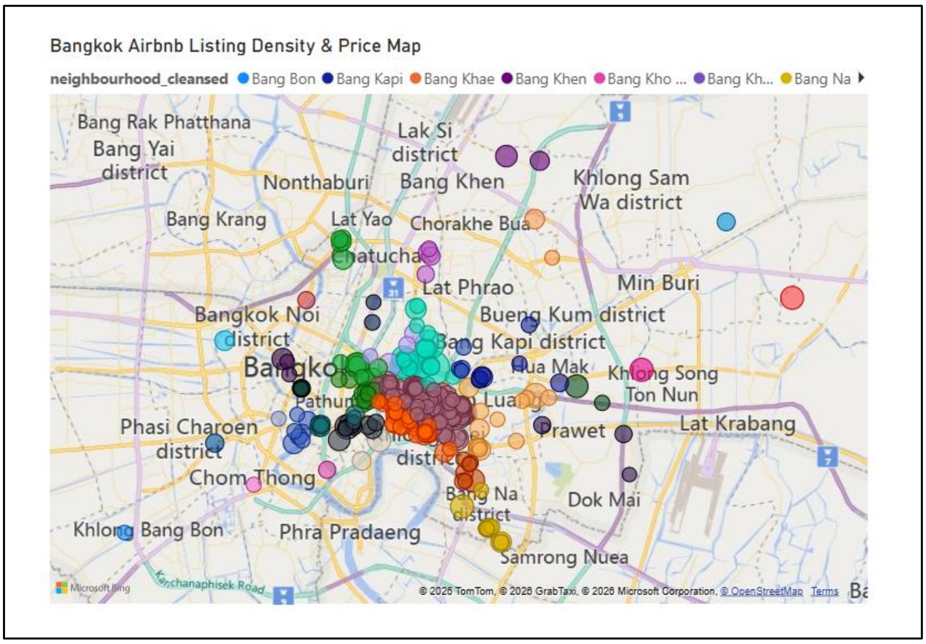



*Figure 23: Power BI Page 5 — Geographic Analysis Map (1)*


*Figure 24: Power BI Page 5 — Geographic Analysis Map (2)*

**Streamlit dashboard:**

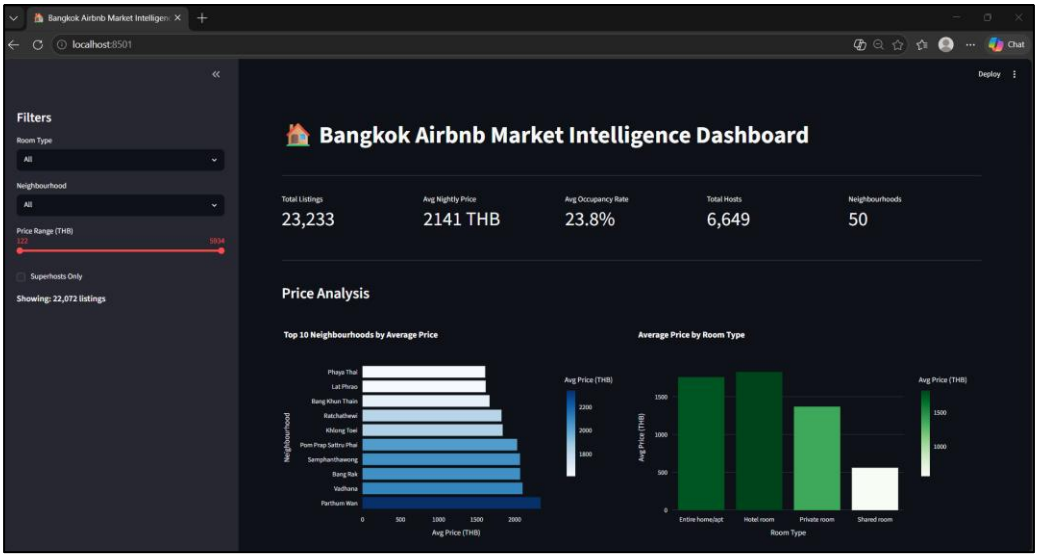



*Figure 25: Streamlit Interactive Market Intelligence Dashboard (1)*

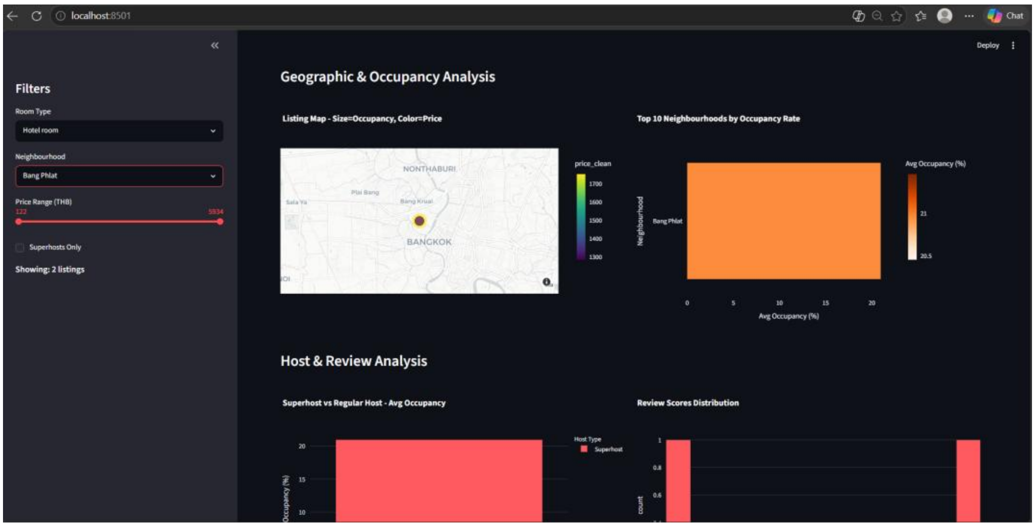



*Figure 26: Streamlit Interactive Market Intelligence Dashboard (2)*

# 11. Business Recommendations

Based on the comprehensive analysis of the Bangkok Airbnb market, the following actionable recommendations are made for three key stakeholder groups: new hosts, existing hosts, and investors.

## 11.1 Recommendations for New Hosts

1. Target Central Bangkok Neighbourhoods

Vadhana, Parthum Wan, and Bang Rak consistently command the highest prices and occupancy rates. New hosts entering the market should prioritize these neighbourhoods when selecting properties. Vadhana alone generated an estimated 1.1 billion THB in total revenue across its listings.

2. List Entire Home Properties

Entire home listings achieve 25.85% occupancy versus 19.04% for private rooms, a 35% advantage. Guests strongly prefer privacy. Where possible, new hosts should list entire properties rather than shared spaces to maximize booking potential.

3. Prioritize Superhost Status Over Premium Pricing

Statistical analysis confirmed that superhosts achieve 50% higher occupancy (30.43% vs 20.21%) despite charging nearly identical prices. New hosts should focus on delivering exceptional guest experiences to achieve superhost status before attempting to command price premiums.

4. Price Around the Median

The market median of 1,378 THB per night is a more reliable pricing benchmark than the mean (2,141 THB), which is distorted by luxury outliers. New hosts should price competitively near the median while building their review base.

## 11.2 Recommendations for Existing Hosts

5. Implement Dynamic Seasonal Pricing

Calendar analysis revealed clear seasonal demand patterns. September shows the lowest availability (46.3%) indicating peak demand, while January to June shows highest availability (70-77%) indicating lower demand. Hosts should raise prices during September and July-August peaks and offer competitive rates during low-demand periods.

6. Reconsider Instant Booking Strategy

Counter to common assumptions, non-instant bookable listings showed higher average occupancy (24.66% vs 22.76%). Hosts who screen guests may be attracting higher-quality, longer-stay bookings. Existing hosts using instant booking should evaluate whether switching to request-based booking improves their booking quality.

7. Focus on Cleanliness and Location Scores

SHAP analysis identified review_scores_location and review_scores_cleanliness as meaningful price predictors. Hosts should actively manage these sub-scores through accurate location descriptions and maintaining high cleanliness standards to support higher pricing.

8. Monitor Sentiment Trends

The declining guest sentiment trend (0.85 in 2011 to 0.52 in 2024) signals rising guest expectations. Existing hosts must continuously improve their service quality to maintain positive reviews as the platform matures and competition intensifies.

## 11.3 Recommendations for Investors

9. Focus on Bedroom Count for Maximum ROI

SHAP analysis identified bedrooms as the single strongest price driver, impacting predictions by 864 THB per bedroom on average. Investors should prioritize properties with more bedrooms over those in slightly better locations, as bedroom count has a larger measurable impact on achievable nightly rates.

10. Watch the Commercial Operator Concentration

Just 2.3% of hosts (commercial operators with 10+ listings) control 25.7% of all supply. This market concentration is increasing as the platform matures. Investors entering the market should consider professional management approaches to compete effectively with these established operators.

11. Bangkok Market Shows Strong Recovery Potential

Review volume grew from 40,567 in 2022 to 148,869 in 2024, a 267% increase in just two years following COVID-19 recovery. This trajectory confirms Bangkok as a high-growth short-term rental market with strong demand fundamentals for continued investment.

# 12. Cross-City Comparisons

This analysis was conducted on a single city - Bangkok, Thailand, rather than multiple cities. This decision was made deliberately based on the assessment's guidance that depth outweighs breadth.

With a 7-day timeline, analyzing multiple cities would have required sacrificing depth across all other sections. The Bangkok dataset alone provided sufficient scale — 603MB of raw data, 10.5 million calendar records, and 583,333 reviews to demonstrate pipeline capability and analytical depth without additional cities.

The pipeline was designed to support additional cities with minimal changes. Adding a new city requires only creating a new data directory and running:

“ run_pipeline(city="london", data_path="data/raw/london") ”

# 13. Limitations & Caveats

## 13.1 Data Limitations

**Single Point-in-Time Snapshot**

The dataset represents a single scrape from September 2025. Price trends, host behavior, and market dynamics cannot be tracked over time from this snapshot alone. Conclusions about trends rely on review dates and calendar projections rather than true historical pricing data.

**Calendar Price Data Unavailable**

Both price and adjusted_price columns in the calendar file were 100% null for Bangkok. This prevented direct analysis of dynamic pricing behavior and seasonal price changes at the daily level. Seasonal insights were derived from availability patterns rather than actual price fluctuations.

**No Actual Booking Data**

Inside Airbnb does not provide actual booking records. Occupancy rates were estimated from calendar availability data, where unavailable dates are assumed to be booked. However, hosts also block dates for personal use, maintenance, or other reasons, meaning occupancy estimates may overstate actual booking rates.

**Review Count as Booking Proxy**

Review volume was used as a proxy for booking demand. However, not all guests leave reviews. Airbnb estimates roughly 70% of guests write reviews. This means actual booking volumes are higher than review counts suggest, and the proxy introduces systematic underestimation of demand.

**Missing License Data**

The license column was 100% null across all 28,806 listings. This prevents any analysis of regulatory compliance or the impact of short-term rental regulations on the Bangkok market.

## 13.2 Analytical Limitations

**Price Outlier Removal**

5,573 listings (19.3%) were removed as extreme price outliers using domain-based thresholds (below 100 THB or above 50,000 THB). While this improves analytical quality, the luxury segment above 50,000 THB is excluded from all pricing analysis and recommendations.

**Median Imputation for Review Scores**

Approximately 35% of listings had missing review scores, imputed using median values. This may underrepresent true variance in review quality for newer listings and could introduce bias in models that use review scores as features.

**English-Only Sentiment Analysis**

VADER sentiment analysis was applied to all reviews regardless of language. Bangkok reviews include significant Thai, Chinese, Japanese, and other language content. Non-English reviews may be incorrectly classified, potentially understating negative sentiment from non-English speaking guests.

**Recommendation System Sample Size**

The listing recommendation system was computed on a sample of 2,000 listings due to memory constraints with cosine similarity matrix computation. A production system would require approximate nearest neighbor algorithms (FAISS, Annoy) to scale to the full 23,233 listing dataset.

## 13.3 Model Limitations

**Price Prediction $R^2$ of 0.51**

The best performing model (Gradient Boosting) explained only 51% of price variance. Key unmodeled factors likely include:

- Specific amenities (pool, gym, parking, WiFi quality)
- Property photos and listing quality
- Host response speed and communication quality
- Proximity to BTS Skytrain stations
- Short-term promotional pricing

**No Temporal Validation**

Models were validated using random train/test splits rather than temporal splits. In a production setting, models should be trained on historical data and validated on future data to simulate real deployment conditions.

### 13.4 Scope Limitations

- Section 3.6 (Advanced Cloud-Native Topics) was not implemented due to time constraints. Architecture discussions are provided in Section 12 as design proposals.
- Cross-city model transfer was not tested.
- Docker containerization was not implemented.
- Unit tests were not written for pipeline components.
- dbt models were not created.

These scope limitations were conscious prioritization decisions made to ensure depth in completed sections rather than superficial coverage of all sections.

## 14. Future Improvements

### 14.1 Data Engineering Improvements

**Incremental Processing**

The current pipeline performs full reprocessing on every run. A production system should implement incremental processing where only newly scraped listings are ingested and processed. This would require change data capture (CDC) logic comparing new scrapes against existing records using listing ID and last_scraped date as change indicators.

**Cloud Migration**

Moving from local DuckDB to a cloud data warehouse such as BigQuery or Snowflake would enable concurrent multi-city processing, team collaboration, and scheduled pipeline runs. The star schema designed in this project maps directly to BigQuery table structures with minimal modification.

**Pipeline Orchestration**

Replacing the current sequential Python pipeline with Apache Airflow or Prefect would provide scheduled execution, dependency management, retry logic, and monitoring dashboards for production deployment.

**Docker Containerization**

Packaging the pipeline in Docker would ensure full reproducibility across different environments. A docker-compose.yml file would allow anyone to run the complete pipeline with a single command without manual environment setup.

**Data Quality Framework**

Implementing Great Expectations or dbt tests would provide automated data quality checks at each pipeline stage, alerting when data distributions shift unexpectedly between scrape cycles.

### 14.2 Analytical Improvements

**Geospatial Feature Engineering**

Adding distance to nearest BTS Skytrain station as a feature would likely improve price prediction significantly. Proximity to transit is a known driver of short-term rental pricing in Bangkok that is not captured by neighbourhood alone.

**Amenity Feature Engineering**

The amenities column contains unstructured text listing all property amenities. Extracting binary flags for high-value amenities (pool, gym, parking, washing machine) and including them as model features could meaningfully improve price prediction $R^2$.

**Multilingual Sentiment Analysis**

Replacing VADER with a multilingual transformer model such as XLM-RoBERTa would provide accurate sentiment analysis across Thai, Chinese, Japanese, and other languages present in Bangkok reviews, giving a more complete picture of guest satisfaction.

**Topic Modeling on Reviews**

Applying LDA or BERTopic topic modeling to review text would surface recurring themes such as location praise, cleanliness complaints, or host communication issues, providing actionable intelligence for hosts beyond simple sentiment scores.

## 14.3 Machine Learning Improvements

**Hyperparameter Tuning**

GridSearchCV or Bayesian optimization could improve Random Forest and Gradient Boosting performance beyond the default hyperparameters used in this assessment.

**Temporal Cross Validation**

Replacing random k-fold validation with time-series cross validation would better simulate real deployment conditions where models are trained on past data and evaluated on future listings.

**Demand Forecasting**

Building a time-series forecasting model on calendar availability data would enable hosts to predict future demand patterns and optimize pricing strategies ahead of peak seasons.

**Cross-City Model Transfer**

Training the price prediction model on Bangkok data and evaluating on another Southeast Asian city such as Singapore or Kuala Lumpur would test model generalization and identify market-specific pricing factors.

## 14.4 Product Improvements

**LLM-Powered Q&A Interface**

The LLM pipeline built in Section 9.3 would be activated with API credits to provide a conversational market intelligence interface. A RAG architecture over the full reviews corpus would improve response accuracy and grounding.

**Real-Time Price Monitoring**

Integrating a streaming architecture using Apache Kafka would enable real-time price monitoring and alerting when competitor listings in the same neighbourhood change pricing, giving hosts actionable competitive intelligence.

**Deployed Streamlit Dashboard**

Deploying the Streamlit dashboard to Streamlit Cloud would make the market intelligence tool publicly accessible without requiring local Python environment setup. This would make the tool immediately usable by non-technical stakeholders.

# 15. Reflection

## 15.1 Prioritization Approach

At the start of this assessment, I read the full brief carefully before writing a single line of code. The design philosophy stated clearly that depth outweighs breadth, and that a candidate completing two sections with exceptional depth will outperform one attempting all sections superficially. This shaped every prioritization decision made throughout the week.

Sections 02, 03, 04, and 05 were treated as the non-negotiable foundation. These cover the core data engineering and analytical competencies the role demands. Only after completing these to a high standard were optional sections 06, 07, and 08 attempted.

## 15.2 Trade-offs Made

| Decision | Trade-off Accepted |
|---|---|
| Single city (Bangkok) | Depth over multi-city breadth |
| DuckDB over cloud tools | Simplicity over production scalability |
| VADER over transformer NLP | Speed over multilingual accuracy |
| Content-based over collaborative filtering | Feasibility over personalization |
| Manual hyperparameters | Time savings over optimal model performance |
| Documented LLM limitation | Transparency over appearance of completeness |

Each trade-off was a conscious decision made with awareness of the consequences rather than an oversight.

## 15.3 What I Would Do Differently

With more time, the following would have been prioritized:

- Amenity feature engineering from the unstructured amenities column would likely have improved price prediction $R^2$ meaningfully
- Multilingual sentiment analysis would have provided more accurate results for the Bangkok market specifically
- Docker containerization would have improved submission reproducibility
- A temporal train/test split would have provided more realistic model validation

## 15.4 Key Lessons Learned

This assessment reinforced several important lessons about real-world data engineering work:

Working with messy real-world data is very different from working with clean tutorial datasets. The Bangkok dataset had 100% null columns, mixed data types, currency symbols in price fields, and ambiguous availability flags. All requiring careful handling and documented decisions.

Statistical significance does not equal practical significance. The hypothesis testing section revealed that with 23,000+ records, almost any difference becomes statistically significant. Effect sizes proved essential for separating meaningful findings from statistical noise.

Transparency about limitations builds more credibility than hiding them. Documenting the LLM pipeline limitation honestly, rather than pretending it worked or omitting it entirely, reflects professional integrity that is more valuable than a polished but incomplete submission.

End-to-end thinking matters. Building a pipeline that flows from raw CSV files through to an interactive dashboard and automated PDF report, with every transformation documented and every decision logged, reflects how production data systems actually work.

### 15.5 Personal Takeaway

This assessment pushed me to think like a data engineer, a statistician, a machine learning practitioner, and a business analyst simultaneously. The most valuable skill I practiced was not any individual technical tool, but the ability to move between these roles fluidly while maintaining a clear focus on delivering business value from raw data.

# Appendix A: AI Usage Disclosure

### A.1 AI Tools Used

| Tool | Version | Purpose |
|---|---|---|
| Claude (Anthropic) | Claude Sonnet | Code assistance, debugging, report guidance |

Claude was the primary AI tool used throughout this assessment. All interactions were conducted through the Claude.ai web interface.

### A.2 AI-Assisted Sections

Section 03 — Data Engineering Pipeline

Claude assisted with the more complex parts of the pipeline including the DuckDB star schema design, the enrichment join logic, and the automated pipeline with logging. I personally wrote and understood the ingestion and profiling scripts, and reviewed every line of the cleaning script to ensure the transformation logic made sense for the Bangkok dataset specifically.

Section 05 — Statistical Analysis

Claude helped me understand which statistical tests were appropriate for non-normally distributed data and assisted with the initial code structure for the Mann-Whitney U and Kruskal-Wallis tests. I independently verified the test selection rationale and rewrote the business interpretations in my own words after reviewing the outputs.

Section 06 — Price Prediction

Claude suggested the overall model comparison structure and helped with the SHAP value implementation, which was new to me. I understood and reviewed the feature engineering decisions and cross-validation logic before including them in the final submission.

Section 07 — AI/ML Experiments

Claude assisted with the TF-IDF recommendation system structure. The sentiment analysis using VADER was largely straightforward and I followed the library documentation directly. The LLM pipeline design was collaborative — Claude helped with the API structure while I designed the business questions and context building logic.

Section 08 — Open Innovation

Claude suggested building a Streamlit dashboard and automated PDF report as the open innovation deliverables. I independently designed the dashboard layout, chose which metrics to highlight, and decided on the visual arrangement. For the Power BI dashboards, Claude suggested which pages to create and recommended axis configurations — but I built every visual myself, made all design decisions, and learned the Power BI interface independently.

Section 12 — Final Report

For the final report, Claude helped me understand the report structure required by the assessment and highlighted the key findings from each section that should be included. Claude also pointed out important technical details such as effect sizes, confidence intervals, and SHAP values, that needed to be explained clearly for non-technical readers.

The actual writing was done by me. I used Claude's guidance on what to include in each section, but wrote the content myself in my own words. Where Claude provided example phrasing or structure, I rewrote it to match my natural writing style. Several sections were written completely independently after Claude outlined what points to cover.

## A.3 Suggestions Rejected or Modified

Not all Claude suggestions were accepted. Notable cases where I disagreed or modified the approach:

- Claude initially suggested analyzing multiple cities. I argued that Bangkok alone provided sufficient depth given the timeline and pushed back on this suggestion. The single-city decision was mine.
- Claude suggested a more complex enrichment join structure initially. I simplified it after reviewing the output and finding the complexity unnecessary for the analytical goals.
- For Power BI, Claude made some axis configuration suggestions that I did not follow when I felt a different arrangement communicated the insight more clearly.

## A.4 Output Validation

All AI-generated code was tested and validated by running it against the actual Bangkok dataset and reviewing outputs for correctness. Where outputs were unexpected (such as the counter-intuitive instant booking finding), I investigated the underlying data directly to confirm the finding was genuine rather than a code error.

Statistical test results were independently verified by checking p-values and effect sizes against expected ranges. Model performance metrics were validated through cross-validation to confirm generalization.

## A.5 Responsible AI Practice

AI tools were used to accelerate implementation of complex components and to learn new techniques quickly. However, intellectual ownership of all analytical decisions, business interpretations, and prioritization choices remained mine throughout. AI was treated as a knowledgeable collaborator rather than an autonomous decision maker.

The LLM pipeline limitation (API credits) was disclosed transparently rather than concealed, consistent with the responsible AI practice of honest documentation of both capabilities and limitations.